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RESEARCH INTERESTS

AI-driven drug discovery; physics-informed machine learning for molecular properties (protonation states, tautomers, pK_a, metabolism); protein–ligand and peptide–protein interaction modeling; structure-based lead optimization; open cheminformatics software and web servers.

RESEARCH HIGHLIGHTS

- **Publications:** 18 peer-reviewed papers, including **6 first-author** papers in *Chem. Sci.* (ChemSci Pick of the Week), *J. Chem. Theory Comput.*, *Chem. Res. Toxicol.*, and *J. Chem. Inf. Model.* (×3).
- **Independent funding:** FWF ESPRIT Fellowship as Principal Investigator (Austrian Science Fund; EUR **349,732**, 2026) and Humboldt Research Fellowship for Postdoctoral Researchers (2026).
- **Prospective validation:** top placements in community blind challenges (CACHE #6 hit discovery; Polaris ADMET 2025 top-5 of 300+; Tox24 5th of 78), plus an experimentally confirmed NMDA-receptor hit.
- **Research software with broad use:** lead developer of MolGpKa, MolTaut, sPhysNet-Taut, and AA-Score; the Xundrug platform has received more than 1,100,000 visits.
- **Academia and industry:** over seven years of research in computer-aided drug discovery, including client-facing project work as Senior Scientist at WuXi AppTec.
- **Mentoring and service:** mentored 5 trainees (3 resulting papers); reviewer for 8 journals including *Chem. Soc. Rev.* and *Chem. Sci.*

PROFESSIONAL APPOINTMENTS AND EDUCATION

Mar. 2024–present **Postdoctoral Fellow, Computational Chemistry**
New York University
Department of Chemistry, New York, NY. Advisor: Prof. Yingkai Zhang.

Jul. 2023–Feb. 2024 **Senior Scientist, Computational Drug Discovery**
WuXi AppTec (Shanghai) Co., Ltd.
Shanghai, China. Client-facing computational drug discovery projects for Kura and Kymera.

Sep. 2018–Jun. 2023 **Ph.D. in Physical Chemistry**
East China Normal University
School of Chemistry and Molecular Engineering, Shanghai, China. Advisors: Profs. John Z. H. Zhang and Changge Ji. Dissertation: *Small Molecule Property Prediction and Protein–Ligand Interaction Calculation.*

Sep. 2014–Jul. 2018 **B.Eng. in Pharmaceutical Engineering**
University of Shanghai for Science and Technology
School of Medical Devices and Food, Shanghai, China.

FELLOWSHIPS, GRANTS, AND AWARDS

2026 **FWF ESPRIT Fellowship (Principal Investigator)**
Austrian Science Fund (FWF), Austria
Total funding: EUR **349,732**. Independent research project hosted by Prof. Johannes Kirchmair, University of Vienna.

2026 **Humboldt Research Fellowship for Postdoctoral Researchers**
Alexander von Humboldt Foundation, Germany
Host: Prof. Paul Czodrowski, Johannes Gutenberg University Mainz.

2025 **NYU Postdoctoral Research and Professional Development Support Grant**
New York University

2022 **Research Excellence Award**
NYU Shanghai
Shanghai, China.

2018 **Outstanding Graduate of Shanghai**
Shanghai, China

2015–2017 **Outstanding Student Scholarship**
University of Shanghai for Science and Technology
Shanghai, China.

BLIND CHALLENGES AND PROSPECTIVE VALIDATION

2025 **CACHE Challenge #6**
Round 1 hit discovery
Discovered a Round 1 hit compound and advanced to Round 2 optimization. Challenge link.

2025 **Drug Discovery 2025 ADMET Challenge**
ASAP Discovery / Polaris
Top-5 placement among more than 300 submissions. Results link.

2024 **Tox24 Challenge**
ICANN2024 / ChemResTox / AIDD / AIChemist
Fifth place among 78 teams; not significantly different from the winning model. Challenge link.

2023 **Shanghai International Computational Biology Innovation Competition**
Drug Screening AI Algorithm "Lingyue" Challenge
Identified an active hit compound targeting the NMDA (N-methyl-D-aspartate) receptor.

PEER-REVIEWED PUBLICATIONS

*Bold: applicant. * Corresponding author. † Trainee mentored or co-mentored by the applicant.*

First-Author Publications

1. **Pan, X.**; Han, C.; Han, F.; Zhang, Y.* Deep Learning of Tautomer Stability from Crystallographic Proton Positions. *Chem. Sci.* 2026. ***ChemSci Pick of the Week, ChemSci Hot Article Collection.***
2. **Pan, X.**; Zhang, X.; Song, X.; Zhang, Y.* Fast and Accurate Prediction of Tautomer Ratios in Aqueous Solution via a Siamese Neural Network. *J. Chem. Theory Comput.* 2025, 21(6), 3132–3141.
3. **Pan, X.**; Gu, Y.; Zhou, W.; Zhang, Y.* Enhancing Transthyretin Binding Affinity Prediction with a Consensus Model: Insights from the Tox24 Challenge. *Chem. Res. Toxicol.* 2025, 38(5), 900–908.
4. **Pan, X.**; Zhao, F.; Zhang, Y.; Wang, X.; Xiao, X.; Zhang, J.*; Ji, C.* MolTaut: A Tool for the Rapid Generation of Favorable Tautomer in Aqueous Solution. *J. Chem. Inf. Model.* 2023, 63(7), 1833–1840.
5. **Pan, X.**; Wang, H.; Zhang, Y.; Wang, X.; Li, C.; Ji, C.*; Zhang, J.* AA-Score: A New Scoring Function Based on Amino Acid-Specific Interaction for Molecular Docking. *J. Chem. Inf. Model.* 2022, 62(10), 2499–2509.
6. **Pan, X.**; Wang, H.; Li, C.; Zhang, J.; Ji, C.* MolGpKa: A Web Server for Small Molecule pK_a Prediction Using a Graph-Convolutional Neural Network. *J. Chem. Inf. Model.* 2021, 61(7), 3159–3165.

Co-Authored Publications

7. Infante, C.†; Lu, J.; **Pan, X.**; Xia, S.; Zhang, Y.* Unifying pK_a and Protonation Prediction with Sequence-Based Deep Learning. *J. Chem. Inf. Model.* 2026, 66(12), 6972–6982.
8. Han, C.; **Pan, X.**; Zhang, Y.* End-to-End Molecular Structure Elucidation from Multimodal NMR Spectra Images Using Vision Transformers. *Chem. Sci.* 2026, 17(26), 13072–13082.
9. MacDermott-Opeskin, H.; et al. (including **Pan, X.**). A Computational Community Blind Challenge on Pan-Coronavirus Drug Discovery Data. *J. Chem. Inf. Model.* 2026, 66(6), 3129–3149.
10. Zhou, W.; Ke, D.; Zhang, Z.; Jin, C.; Wu, Y.; **Pan, X.**; Wang, X.; Xiao, X.; Zhang, K.*; Ji, C.* qMol: A Web

Server for Efficient Molecular Queries Using Fragment-Based Reduced Graphs. *J. Chem. Inf. Model.* 2026, 66(1), 7–15.

11. Ke, D.; Zhou, W.; Zhang, Z.; Jin, C.; Wu, Y.; **Pan, X.**; Wang, X.; Xiao, X.; Ji, C.* SiteMatcher: A Web Server for Structure-Based Drug Design Using Protein–Ligand Interaction Patterns. *J. Chem. Inf. Model.* 2025, 65(24), 13028–13036.
12. Zhang, Z.; Ke, D.; Jin, C.; Zhou, W.; **Pan, X.**; Zhang, Y.; Wang, X.; Xiao, X.; Ji, C.* MacGen: A Web Server for Structure-Based Macrocycle Design. *J. Chem. Inf. Model.* 2024, 64(24), 9048–9055.
13. Zhang, Y.†; Zhang, Z.; Ke, D.; **Pan, X.**; Wang, X.; Xiao, X.; Ji, C.* FragGrow: A Web Server for Structure-Based Drug Design by Fragment Growing within Constraints. *J. Chem. Inf. Model.* 2024, 64(10), 3970–3976.
14. Wang, H.†; **Pan, X.**; Zhang, Y.; Wang, X.; Xiao, X.; Ji, C.* MolHyb: A Web Server for Structure-Based Drug Design by Molecular Hybridization. *J. Chem. Inf. Model.* 2022, 62(12), 2916–2922.
15. Wei, M.; Zhang, X.; **Pan, X.**; Wang, B.; Ji, C.; Qi, Y.*; Zhang, J.* HobPre: Accurate Prediction of Human Oral Bioavailability for Small Molecules. *J. Cheminf.* 2022, 14(1), 1–10.
16. Shao, G.; Bao, J.; **Pan, X.**; He, X.*; Qi, Y.*; Zhang, J.* Computational Analysis of Residue-Specific Binding Free Energies of Androgen Receptor to Ligands. *Front. Mol. Biosci.* 2021, 8, 36.
17. Shao, G.; Bao, J.; **Pan, X.**; He, X.*; Qi, Y.*; Zhang, J.* Analysis of the Binding Modes of the First- and Second-Generation Antiandrogens with Respect to F876L Mutation. *Chem. Biol. Drug Des.* 2021, 98(1), 60–72.
18. Shan, J.; **Pan, X.**; Wang, X.; Xiao, X.; Ji, C.* FragRep: A Web Server for Structure-Based Drug Design by Fragment Replacement. *J. Chem. Inf. Model.* 2020, 60(12), 5900–5906.

SOFTWARE AND WEB SERVERS

- **Xundrug web-server platform** (xundrug.cn): Python-based platform hosting computational chemistry and drug-discovery tools developed by the ECNU-NYUSH computational chemistry team; more than 1,100,000 visits.
- **Molecular property prediction (lead developer):** MolGpKa for small-molecule pK_a prediction; MolTaut for favorable tautomer generation; sPhysNet-Taut for tautomer stability and tautomer-ratio prediction.
- **Protein–ligand scoring (lead developer):** AA-Score, an amino-acid-specific scoring function for rapid estimation of protein–ligand binding affinity.
- **Structure-based design web servers (co-developer):** SiteMatcher, qMol, MacGen, FragGrow, FragRep, and MolHyb for binding-site matching, molecular query, macrocycle design, fragment growing, fragment replacement, and molecular hybridization.

MENTORING AND SUPERVISION

Mentored or co-mentored five trainees (two Ph.D., two M.Sc., one undergraduate); three trainee-led projects have been published.

Mar. 2025–present	**Seayoung Lee** *Ph.D. student*	Supervise a drug discovery project to identify novel inhibitors of WaaL O-antigen ligase; guide data collection, model development, result analysis, coding, and project planning.
Jun. 2024–present	**Charlotte Infante** *Ph.D. student*	Co-mentor a T5Chem language-model project for pK_a and protonation prediction; advise model design, benchmarking, interpretation, and manuscript preparation. *Outcome: J. Chem. Inf. Model.* 2026.
Sep. 2022–Jun. 2023	**Xiaoyu Huang** *B.Sc. student*	Led a bachelor's thesis project on binding-site matching; guided descriptor design, database construction, programming, and thesis writing.
Sep. 2021–Jun. 2023	**Yueing Zhang** *M.Sc. student*	Guided a structure-based fragment-growing project, including fragment database construction, descriptor design, coding, and research communication. *Outcome:* FragGrow, *J. Chem. Inf. Model.* 2024.

Sep. 2020–Jun. 2022	**Hao Wang** *M.Sc. student*	Directed a molecular-hybridization project for structure-based drug design; trained the student in RDKit workflows, implementation, analysis, and manuscript writing. *Outcome:* MolHyb, *J. Chem. Inf. Model.* 2022.

PRESENTATIONS AND CONFERENCES

Mar. 2026	**ACS Spring 2026** *Atlanta, GA* Oral presentation.
Jun. 2025	**NYC AI4Chemistry Summit 2025** *New York, NY* Poster presentation.
Mar. 2025	**ACS Spring 2025** *San Diego, CA* Poster presentation.
Sep. 2024	**Simons Center Postdoctoral Research Symposium** *New York, NY* Oral presentation.
Nov. 2019	**15th National Conference on Computer Chemistry of China** *China* Poster presentation.
2018–2023	**Weekly Seminar, Center for ECNU-NYUSH Computational Chemistry** *Shanghai, China* Regular research presentations and discussion.

PROFESSIONAL SERVICE

Journal reviewer: *Chemical Society Reviews*; *Chemical Science*; *Journal of Chemical Information and Modeling*; *Journal of Cheminformatics*; *Journal of Computational Chemistry*; *Chemical Research in Toxicology*; *Journal of Computer-Aided Molecular Design*; *Reaction Chemistry & Engineering*.

TECHNICAL SKILLS

Programming	Python, PyTorch, C++, SQL, shell scripting, Linux, RDKit.
Computational tools	Schrödinger, OpenMM, Gaussian, AMBER, PySCF, gpu4pyscf.
Methods	Deep learning, cheminformatics, quantum chemistry, molecular docking, virtual screening, protein–ligand scoring, molecular-property prediction, web-server development.